ARMAAN SINHA

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SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning with strong foundations in Python, Data Structures & Algorithms, and Software Engineering. Experienced in building and deploying machine learning models and full-stack web applications. IEEE-published researcher seeking Software Engineering / Machine Learning Internship roles.

EDUCATION

Manipal University Jaipur, *Jaipur, Rajasthan, India* Aug 2023 – May 2027
B.Tech., Computer Science & Engineering (AI & ML) | CGPA: 7.79 / 10
Relevant Coursework: Deep Learning · Machine Learning · Data Structures & Algorithms · DBMS · Operating Systems · Quantum Computing · OOP · Computer Networks

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS, JavaScript
AI / ML: Machine Learning, Deep Learning, Reinforcement Learning, MARL, CNN, VGG16, Random Forest, XGBoost, Gradient Boosting, SVM, KNN, Logistic Regression, Feature Engineering, Transfer Learning.
Frameworks & Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Flask, Django, Streamlit, Node.js
Tools & Environments: Git, GitHub, Linux, REST APIs, PettingZoo, OpenAI Gym, Qiskit, Jupyter Notebook, Google Colab
Soft Skills: Analytical Thinking, Problem Solving, Research & Documentation, Collaborative Teamwork, Software Development Life Cycle(SDLC), Agile Development, Attention to Detail, Technical Communication, Self-Learning

PROJECTS

Behavioral Observatory Framework – Failure Detection in MARL Jan 2026 – Apr 2026
Python · Random Forest · PettingZoo · Scikit-learn · Multi-Agent Reinforcement Learning

- Designed a non-intrusive monitoring framework to detect and diagnose failures in MARL systems; engineered a 28-dimensional feature vector capturing observation, action, reward, and temporal dynamics from agent episode traces.
- **Achieved 94.7% binary detection accuracy (F1: 0.93)** and 84.2% multi-class accuracy (F1: 0.84) across 18 failure types in 5 categories using a Random Forest classifier.
- Generated 5,700 synthetic episodes via controlled failure injection in PettingZoo's simple_spread environment; benchmarked against SVM, Gradient Boosting, KNN, and Logistic Regression — validating Random Forest superiority.

VibeMatch – Emotion-Aware Recommendation System Jan 2025 – Nov 2025
Python · CNN · VGG16 · Flask · Scikit-learn · REST APIs · Deep Learning

- Trained a CNN (VGG16) on FER-2013 (35,000+ images) via transfer learning for real-time facial emotion recognition; integrated Spotify, IMDB, and Google Books APIs through a Flask backend for personalized recommendations.
- **Published peer-reviewed paper at IEEE ISMAC-2025** on CNN-based emotion-aware recommender systems; recognized for novel application of computer vision to recommendation engines.

Employee Attrition Prediction System Sept 2025 – Dec 2025
Python · XGBoost · Random Forest · Flask · Streamlit · Scikit-learn

- Built end-to-end ML pipeline — EDA, feature engineering, model training — benchmarking Logistic Regression, Random Forest, and XGBoost; improved recall for high-risk employees by ~18% through hyperparameter tuning and class-balancing.
- Deployed REST-based web application via Flask + Streamlit, enabling real-time HR decision support with an interactive prediction interface.

CERTIFICATIONS & PUBLICATIONS

- **Certifications:** OOP Using Python – Cisco · CCNA Switching, Routing & Wireless – Cisco · Database Programming with SQL – Oracle · Software Project Management – Google (Coursera)
- **Publication:** "VibeMatch: CNN-Based Emotion-Aware Recommendation System," IEEE ISMAC-2025.